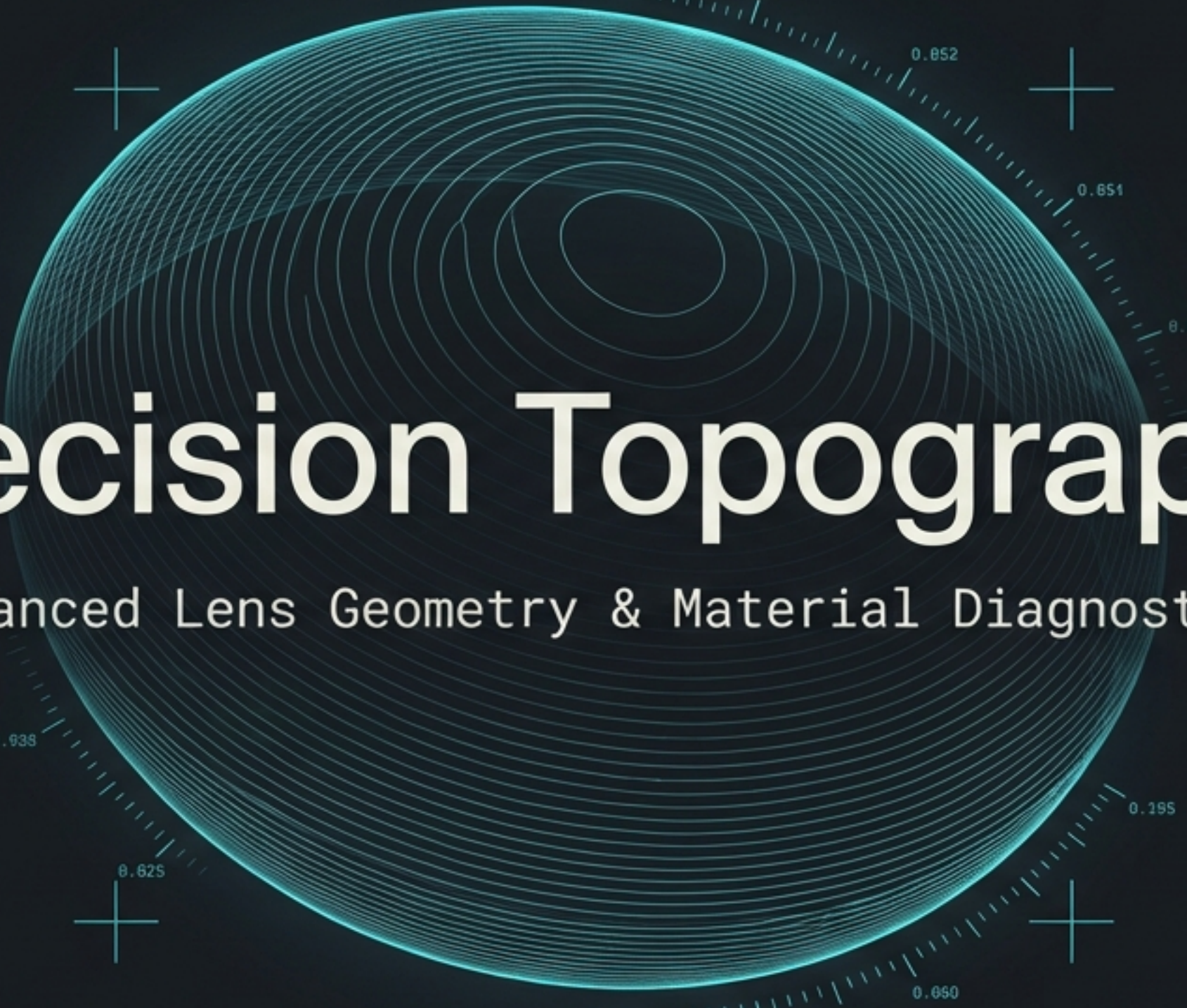


A large, glowing cyan sphere with concentric rings, resembling a lens or a topographical map, is the central focus. It is surrounded by a dark blue background with a grid and faint topographical lines. Four white crosshair markers are positioned around the sphere. A curved scale with numerical values is visible in the upper right and lower right areas.

Precision Topography

Advanced Lens Geometry & Material Diagnostics

The Topographical Toolkit

Geneva Lens Clock

Base Curve Verification &
Toric Mapping

Analog Dioptric Dial

Mechanical Needle Fluctuation

Fixed Index
(Typically 1.53)

Primary
Diagnostic
Function

Interface
Output

Cylinder
Detection

Calibration
Baseline

Digital Lens Clock

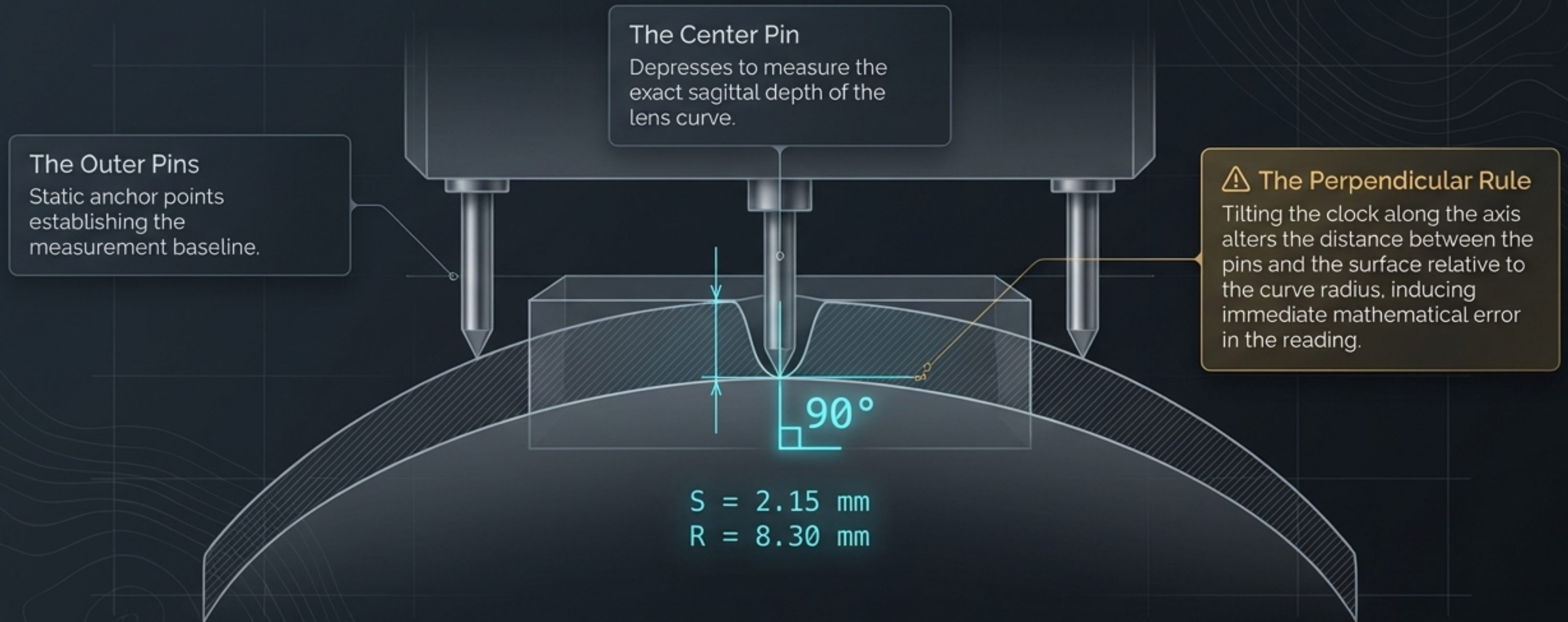
Refractive Index Calculation

LCD Screen (+/- Data Readout)

Multi-Axis Axis Plotting

Variable/Selectable
(1.49 to 1.74)

The Geometry of Sagittal Measurement



Analog Profiling: The Base Curve



1. Isolate the Front Surface

Place the clock perfectly flush against the anterior curvature of the lens.

2. Read the Primary Power

Record the primary spherical dioptric power indicated by the needle.

3. Clinical Verification

Compare this base curve to the patient's previous eyewear to rule out base curve changes as the culprit for new visual discomfort or non-adaptation.

Mechanical Detection of Plus Cylinder Designs



BASE: 6.00 D

Align & Rest

The clock sits on the front surface. The needle registers a base power.



READING: 6.75 D

Rotate on Axis

The clock is smoothly rotated 360 degrees around the optical center.

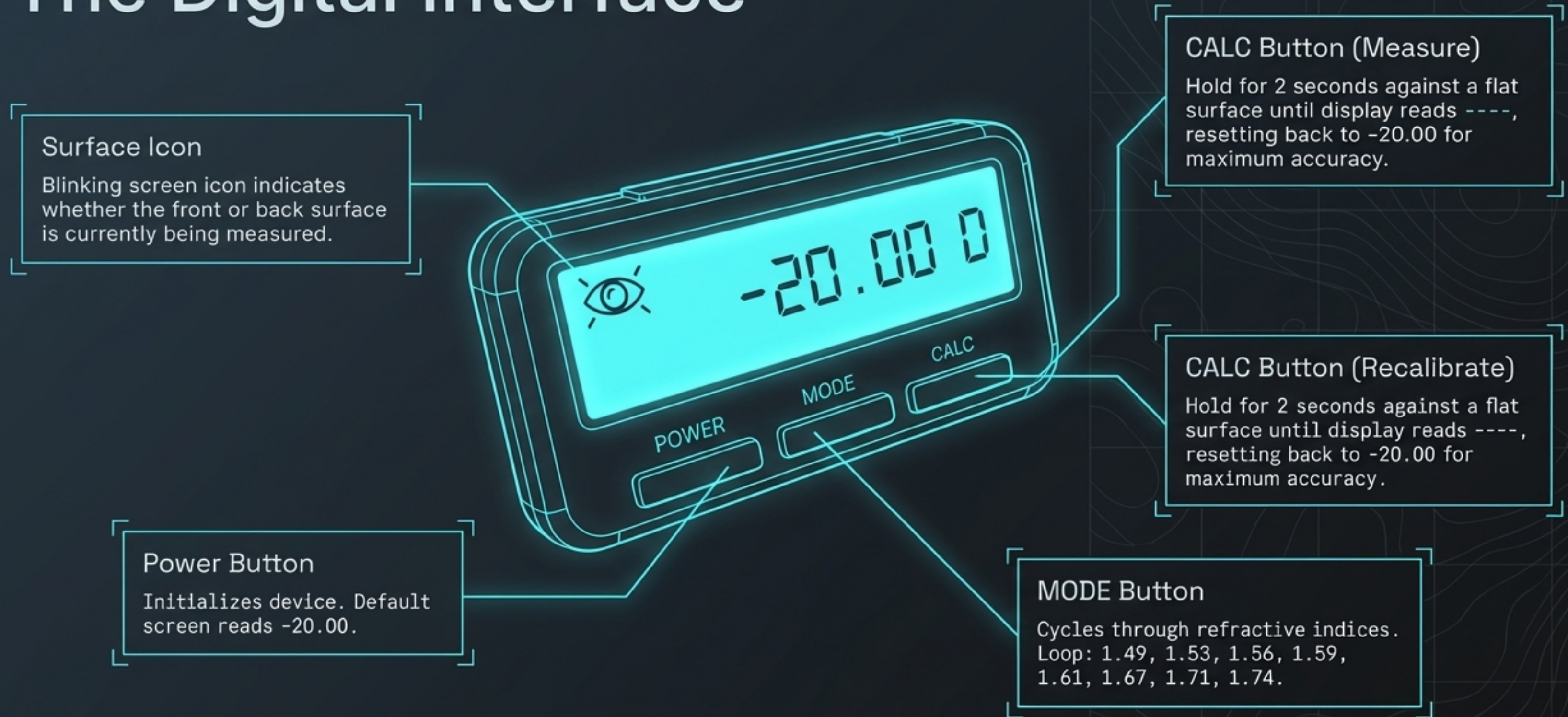


MAX: 7.50 D

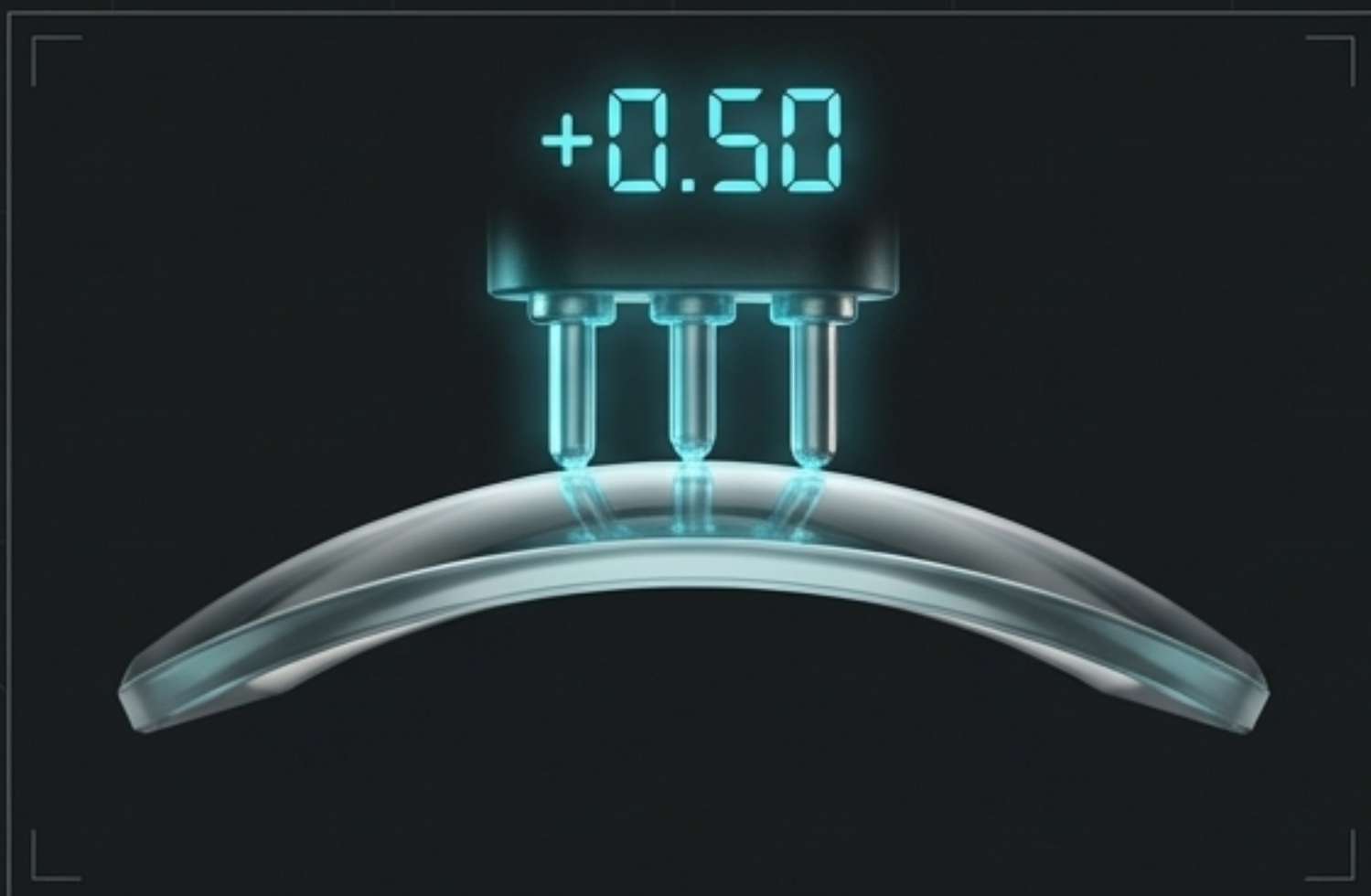
Observe Fluctuation

If the needle remains static, the surface is spherical. If the needle fluctuates between a minimum and maximum reading, the surface is toric. The highest reading represents the steepest meridian, confirming a plus cylinder design.

The Digital Interface



Capturing Total Dioptric Curvature



Step 1: Set the Target Index

Use **MODE** to lock in the suspected refractive index of the material.

Step 2: Anterior Measurement

Press **CALC** on the front surface. Note the blinking icon confirming the front reading.



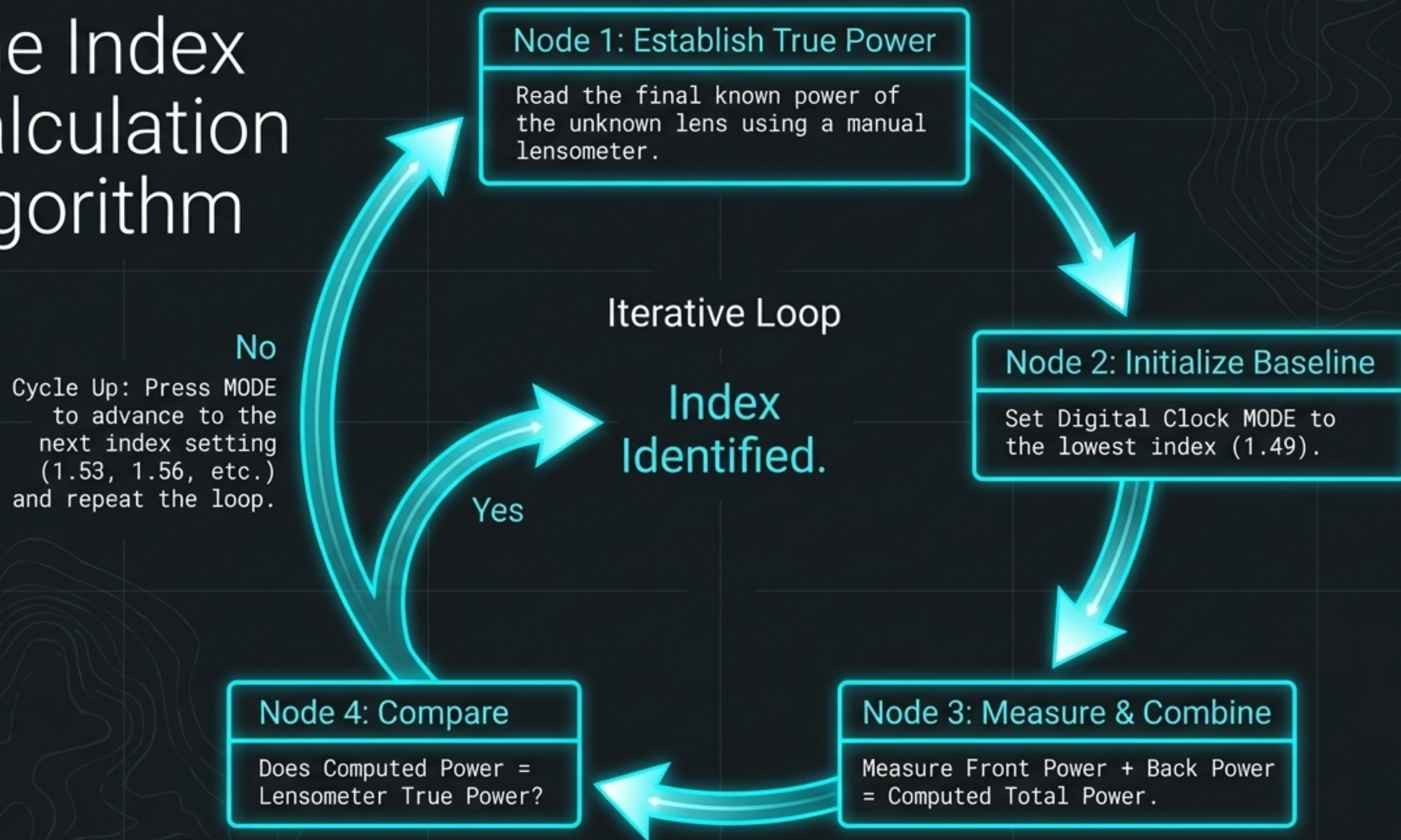
Step 3: Axis Alignment

Carefully flip the lens, orienting the pins in the exact same approximate axis and location.

Step 4: Posterior Measurement

Press **CALC** on the back surface. The device utilizes these two localized curves to compute total surface powers.

The Index Calculation Algorithm



Index Iteration Case Study

TARGET GOAL: -5.00D

Context: Lensometer True Power = -5.00D. Material is unknown (non-polycarbonate).

Baseline Attempt

Setting: 1.49 Index

$$\begin{array}{r} \text{Front } (+0.25) \\ + \text{ Back } (-3.50) \\ \hline = -3.25\text{D} \end{array}$$

Result: Mismatch. Cycle up past 1.53 and 1.56 based on variance.

Refining

Setting: 1.59 Index

$$\begin{array}{r} \text{Front } (+0.50) \\ + \text{ Back } (-5.25) \\ \hline = -4.75\text{D} \end{array}$$

Result: Close, but cycle up one more step to be sure.

Confirmation

Setting: 1.61 Index

$$\begin{array}{r} \text{Front } (+0.50) \\ + \text{ Back } (-5.50) \\ \hline = -5.00\text{D} \end{array}$$

Result: Match. High probability the material is a 1.61 High-Index lens.

The Complete Investigative Workflow



True optical mastery requires integrating overall refractive power with precise surface geometry and material science to solve complex clinical mysteries.